

Tensor computations

Comon's exact-rank conjecture for Hankel tensors

Difficulty: challenging **Importance:** interesting to specialist**Status:** Solved

Rating rationale: Challenging because exceptional Hankel tensors require a uniform comparison of structured and unrestricted exact decompositions; specialist importance reflects this particular structured tensor class.

Resolution: affirmative, 12 September 2026

Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, proves the complete displayed target in [Theorem 1.1, Sections 2–5](#): ordinary and symmetric rank agree for every complex Hankel tensor of every order $m \geq 3$ and dimension $n \geq 2$, including zero and exceptional tensors. No genericity or Vandermonde restriction is imposed. The stronger formula for nonzero tensors is

$$R(H) = R_{\text{sym}}(H) = \min\{D - r + 2, (m - 1)r - (m - 2)s\}, \quad D = m(n - 1),$$

where r is the middle Hankel catalecticant rank and s counts distinct projective roots of a smallest-degree binary apolar polynomial. The balanced case is independent of its choice. [Proof source](#) · [Submission, provenance and verified affiliation](#).

The complete argument passed a separate [independent Codex AI-agent audit](#). This is informal automated review, not external human peer review or formal verification. No Lean verification was performed. The original target and prior-source credit are retained below; ratings above are historical assessments of that target.

Statement

For every $m \geq 3$, $n \geq 2$, and $h \in \mathbb{C}^{m(n-1)+1}$, form the symmetric tensor

$$H_{i_1 \dots i_m} = h_{i_1 + \dots + i_m - m}, \quad 1 \leq i_j \leq n.$$

Is $R(H) = R_{\text{sym}}(H)$ always true? Here $R(H)$ is the least integer $r \geq 0$ admitting

$$H = \sum_{j=1}^r u_{j,1} \otimes \dots \otimes u_{j,m}, \quad u_{j,k} \in \mathbb{C}^n,$$

whereas $R_{\text{sym}}(H)$ is the least r admitting $H = \sum_{j=1}^r c_j v_j^{\otimes m}$, with $c_j \in \mathbb{C}$ and $v_j \in \mathbb{C}^n$. Empty sums represent zero. No genericity or Vandermonde restriction is imposed.

Relevance

This would justify symmetry-preserving decomposition lengths for every Hankel tensor, including exceptional data. Such structure appears in harmonic retrieval and structured tensor factorization.

References

1. J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM J. Matrix Anal. Appl. 40 (2019). [DOI; primary preprint](#), §6, Conjecture 6.3, p.16.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. [Primary preprint](#), §1, equation (1), and §4, Theorem 3, for Hankel structure and decomposition.

Status check — 2026-09-10

Rechecked [Nie–Ye, Conjecture 6.3, Corollary 4.3 and Theorem 5.4](#), and searched for later Hankel Comon proofs and counterexamples. Generic even-order and order-three Hankel tensors are proved cases of the displayed universal equality; exceptional tensors and the remaining orders are not settled by those results. The paper already accounts for Shitov’s unrestricted counterexample. No later full resolution was located.